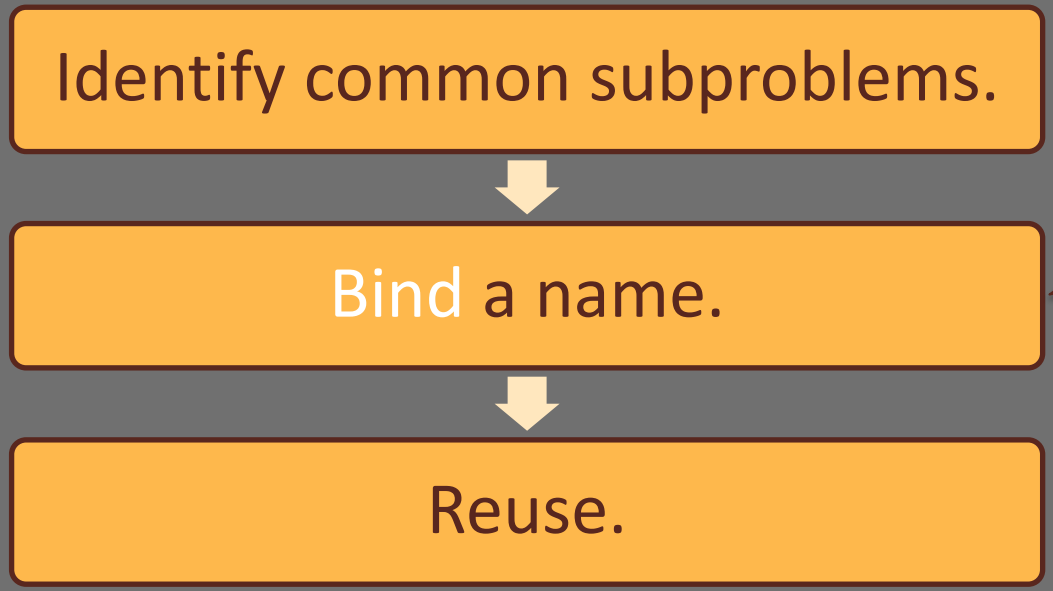
Binding Forms
(aka Simplifying Complex Queries)

Code from previous exercise

Why is repeating code bad?

```
SELECT crn
FROM (SELECT crn, Avg(rating) AS avg_rating1
      FROM Enrollment
      GROUP BY crn
     ) AS Average1
WHERE avg_rating1 = (SELECT Max(avg_rating2)
                     FROM (SELECT Avg(rating) AS avg_rating2
                           FROM Enrollment
                           GROUP BY crn
                          ) AS Average2
                    );
```

Standard practice



SQL:

- WITH
- Temporary table

WITH – Using in one query

Common Table Expression (CTE)

```
WITH Average AS
    (SELECT crn, Avg(rating) AS avg_rating
     FROM Enrollment
     GROUP BY crn)
SELECT crn
FROM Average
WHERE avg_rating = (SELECT Max(avg_rating)
                    FROM Average);
```

Preferred syntax

```
WITH Average (crn, avg_rating) AS
     (SELECT crn, Avg(rating)
      FROM Enrollment
      GROUP BY crn)
SELECT crn
FROM Average
WHERE avg_rating = (SELECT Max(avg_rating)
                    FROM Average);
```

Temp table – Using in multiple queries

Automatically dropped at end of session.

```
CREATE TEMPORARY TABLE MyTable (
    …);
```

```
SELECT INTO TEMPORARY TABLE MyTable
…;
```